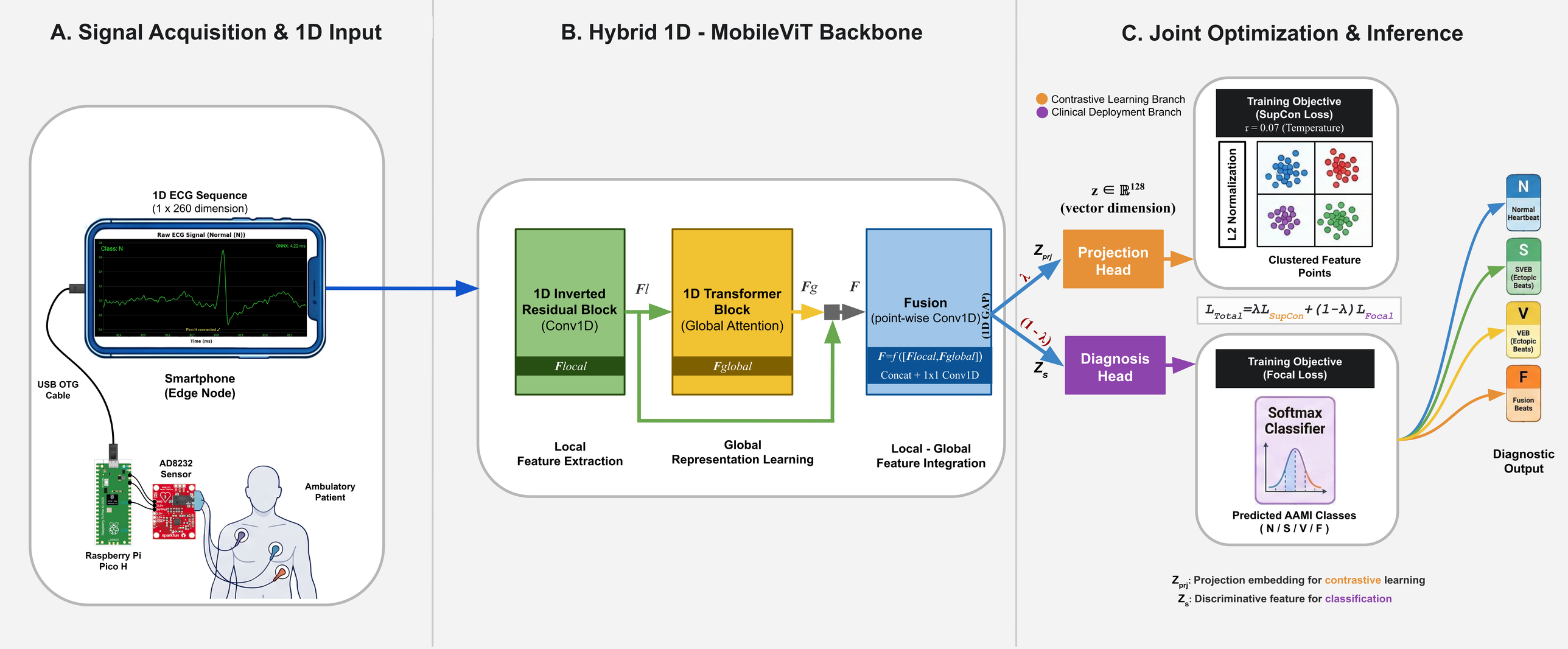

A. Signal Acquisition & 1D Input
1D ECG Sequence
(1 x 260 dimension)
Raw ECG Signal (Normal (N))
Class: N
Time (ms)
Smartphone
(Edge Node)
USB OTG
Cable
AD8232
Sensor
Raspberry Pi
Pico H
Ambulatory
Patient
B. Hybrid 1D - MobileViT Backbone
1D Inverted
Residual Block
(Conv1D)
Flocal
Fl
1D Transformer
Block
(Global Attention)
Fglobal
Fg
F
Fusion
(point-wise Conv1D)
(1D GAP)
F=f([Flocal,Fglobal])
Concat + 1x1 Conv1D
Local
Feature Extraction
Global
Representation Learning
Local - Global
Feature Integration
C. Joint Optimization & Inference
Contrastive Learning Branch
Clinical Deployment Branch
z ∈ ℝ128
(vector dimension)
Z prj
λ
Projection
Head
Training Objective
(SupCon Loss)
τ = 0.07 (Temperature)
L2 Normalization
Clustered Feature
Points
L Total=λL SupCon+(1−λ)L Focal
(1 - λ)
Z s
Diagnosis
Head
Training Objective
(Focal Loss)
Softmax
Classifier
Predicted AAMI Classes
(N / S / V / F)
N
Normal
Heartbeat
S
SVEB
(Ectopic
Beats)
V
VEB
(Ectopic
Beats)
F
Fusion
Beats
Diagnostic
Output
Z prj: Projection embedding for contrastive learning
Z s: Discriminative feature for classification